shortened version of his thesis—with a new title, "'Relative State' Formulation of Quantum Mechanics"—which emphasized the mathematical formalism of the universal wave function and downplayed the "splitting" into many worlds. With Wheeler's approval, Everett finally received his PhD in physics from Princeton in April 1957. His shortened thesis was judged "very good" and published in *Reviews of Modern Physics*. It appeared with a short companion paper by Wheeler, where he claimed that Everett's interpretation "does not seek to supplant the [Copenhagen interpretation], but to give a new and independent foundation for that [interpretation]."

Nonetheless, the physicists in Copenhagen still didn't agree with Wheeler. Everett has "some confusion as regards the observational problem," wrote Bohr to Wheeler after Wheeler sent him a copy of Everett's shortened thesis. Bohr, true to form, said he didn't have time to write down all of his ideas on the subject and promised that Petersen would write with a more detailed reply to Everett. Petersen's comments were indeed more extensive, and more devastating. "I think that most of us here [in Copenhagen] look differently upon the problems and don't feel those difficulties in quantum mechanics which your paper sets out to remove," Petersen wrote. "The very idea of observation belongs to the frame of classical concepts." In other words, Petersen and the others in Copenhagen thought that the process of observation had to be classical—that it was impossible in principle to explain observations using quantum physics. Instead, the world must be split in two: the classical and the quantum, and quantum physics could never be used to describe classical events like observations and measurements. But several sentences later in the same letter, Petersen contradicts himself—he says that there are quantum effects in measurement devices, but they can be safely ignored because the devices are large. Astonishingly, Petersen uses this to justify the split between the classical and quantum, the same split that supposedly makes it impossible to give a quantum description of a measurement apparatus in the first place! "There is no arbitrary distinction between the use of classical concepts and the [quantum] formalism since the large mass of the [measurement] apparatus compared with that of the individual atomic objects permits that neglect of quantum effects," Petersen wrote. Everett spotted the contradiction immediately and called Petersen on it in his reply. "You talk